EXCELLENCE

CLASS 11 - 2026-27

Physics | Units 01

Units and Measurement

Physical Quantities, Dimensions and Conversion

By - Excellence Programme

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Topics to be covered

01 Physical quantity and why we measure

02 Fundamental, derived, supplementary

03 SI units and how to write them (CBSE)

04 Dimension — simple meaning

05 Must-know dimensions (v, a, F, W, P)

06 Unit conversion method

07 New-system of units

08 Board practice

Where we are today

Basic Maths and Vectors are complete. First NCERT physics chapter starts now.

This class merges PW Units 01 and Units 02 into one teaching hour.

Physical quantity → dimension → unit conversion

Do not memorise later-chapter formulas today. Learn the method.

Units and Measurement

1. Type of physical quantity and unit conversion (today)
2. Dimensional analysis
3. Significant figures
4. Errors
5. Measuring instruments

If a later chapter formula appears, do not panic. The method is the same.

Physical quantity

Physical quantity = a quantity that can be measured

Examples: mass, length, time, force, energy.

A measurement is written as n u (number × unit).

Beauty, intelligence, honesty are not physical quantities.

Rough work / extra work

Need for measurement (2 marks)

Write this in simple English. Do not make it long.

1. Physics is the study of nature using numbers.

2. To compare two things, we must measure them.

3. Measurement = how many × a standard unit fits into the quantity.

Physical quantity = numerical value x unit

Always write the unit. A number without a unit is incomplete.

How to write any 3-mark theory answer

Step 1: Definition in one line (easy words).

Step 2: Formula or symbol, if any.

Step 3: One example.

Step 4: One extra point (use / difference / trap).

Definition + formula + example = full marks

Do not copy hard English. Examiner wants clear points, not big words.

Three types of physical quantity

Fundamental

Independent. All other quantities are built from these.

Derived

Made from fundamentals: v = s/t, F = m a.

Supplementary

Angle (rad) and solid angle (sr). Have a unit, no dimension.

Fundamental means: you cannot derive one of them from the others.

Two quantities can share a unit and still be different (work and torque).

Fundamental vs derived (3 marks)

Point	Fundamental	Derived
Meaning	Independent	Made from fundamentals
Count in SI	Seven only	Many
Example	mass, time	speed, force
Unit	kg, s, m ...	m/s, N, J ...
Can we skip one?	No	Yes, if defined

End the answer with: All derived quantities are built from the seven base ones.

Seven SI base quantities

Quantity	SI unit	Symbol
Mass	kilogram	kg
Length	metre	m
Time	second	s
Temperature	kelvin	K
Current	ampere	A
Amount of substance	mole	mol
Luminous intensity	candela	cd

Rough work / extra work

Systems of units (2–3 marks)

CGS: centimetre, gram, second.

FPS: foot, pound, second. MKS: metre, kilogram, second.

SI: international system. It starts from MKS and adds A, K, mol, cd.

We use SI because it is one system for the whole world.

Advantage: same unit everywhere, easy to convert, coherent (no extra numbers).

Write SI, not S.I. in running text. Say 'SI unit of force is newton (N)'.

Rules for writing units (copy this)

1. Unit name is not capital: metre, newton, joule. (Exception: Celsius.)

2. Symbol may be capital if named after a person: N, J, W, K.

3. Never write plural: 10 kg, not 10 kgs. Do not put a full stop after the symbol.

4. Leave a gap: 5 m, 10 N. Write m/s or m s^{-1}, not mps.

5. Quantity, unit and symbol are three different things.

Quantity: force Unit: newton Symbol: N

Wrong: 10 Newton / 10 n / 10 nts. Right: 10 N.

Derived quantities from definition

Velocity = displacement / time → m/s

Acceleration = change in velocity / time → m/s^2

Force = mass x acceleration → kg m/s^2 = 1 N

Momentum = mass x velocity → kg m/s

Energy = (1/2) m v^2 → kg m^2/s^2 = 1 J

Power = energy / time → J/s = W

Same unit does not prove same physical quantity.

Which group can be fundamental?

Which of the following groups may be treated as a set of fundamental quantities?

1. mass, momentum, velocity
2. force, acceleration, mass
3. force, time, velocity
4. none of these

Solution: Which group can be fundamental?

Step 1: Fundamental means independent: none can be written using only the others.

Step 2: (a) p = m v, so momentum is derived. (b) F = m a, so they are linked.

Step 3: (c) F, t, v cannot make each other. They can form a valid base set.

Answer: force, time, velocity.

Rough work / extra work

Characteristic of a unit

Which of the following is a characteristic of a unit?

1. universally accepted
2. invariable and well defined
3. suitable size and easily available
4. all of the above

Solution: Characteristic of a unit

Answer: all of the above.

A good unit — 3 points to remember

Write these three lines. That is the full 3-mark answer.

1. It must be the same for everyone (universally accepted).

2. It must not change with time or place (invariable, well defined).

3. It should be of handy size and easy to copy (easily available).

Accepted + does not change + easy to use

Hand-span and foot-step failed because they change from person to person.

What is a dimension?

Dimension = power of a base quantity in the expression of a P.Q.

Mass → M^1 Length → L^1 Time → T^1 Current → A^1

Temperature → K^1 (or θ^1) Amount → mol Luminous intensity → cd

The letter M is mass, not metre. Metre is the unit of length L.

Dimensional formula — how to say it

Dimensional formula: the expression that shows powers of M, L, T.

Example: [F] = M L T^{-2}. This is the dimensional formula of force.

Dimensional equation: quantity = its dimensional formula.

Example: [force] = [M L T^{-2}].

Write [ ] around the quantity. Then write M, L, T with powers.

CBSE likes the word 'dimensional formula'. Use it, not only 'dimension'.

Rough work / extra work

Unit vs dimension (2 marks)

	Unit	Dimension
Meaning	standard used to measure	powers of base quantities
Can change?	yes (m or cm)	no (length is always L)
Example	newton, joule	$M\ L\ T^{-2}$
Depends on system?	yes (SI / CGS)	no

Force in SI is N, in CGS is dyne. Dimension is $M\ L\ T^{-2}$ in both.

Dimensionless quantities (2 marks)

A quantity with all powers zero is dimensionless: $M^0 L^0 T^0$.

It may still have a unit. Angle has unit radian, but no dimension.

Examples: angle, solid angle, strain, refractive index, relative density.

No dimension does not always mean no unit

Strain = change in length / original length = L/L = 1. Pure number.

Anything that is a length

Displacement, distance, height, width, diameter, amplitude,

wavelength, radius of curvature, range of projectile, mean free path,

focal length, circumference, fringe width, position, elongation.

All of these have dimension $L^1\ M^0\ T^0$

If it is a measured length, write [L]. Do not overthink the name.

Velocity family vs angular velocity

$$[\text{velocity}] = L\,T^{-1}$$

Same dimension: speed, average / orbital / escape / terminal / drift velocity.

$$[\text{angular velocity } \omega] = T^{-1} \quad \text{because } \omega = \theta / t$$

ω is not a velocity. v = r ω mixes L into it.

Mechanics dimensions you must own

Quantity	From	Dimension
Acceleration	v / t	$L\,T^{-2}$
Force	m a	$M\,L\,T^{-2}$
Momentum / impulse	m v	$M\,L\,T^{-1}$
Work / energy / torque	F s	$M\,L^{2}\,T^{-2}$
Power	W / t	$M\,L^{2}$ T^(-3)
Pressure / stress	F / A	$M\,L^{-1}\,T^{-2}$
Density	m / V	M L^(-3)

Work and torque share dimension. They are still different quantities.

Rough work / extra work

Angular set and moment of inertia

Angle θ has no dimension: $M^0\ L^0\ T^0$. Unit is radian.

Angular velocity $\omega = \theta / t \rightarrow T^{-1}$

Angular acceleration $= \omega / t \rightarrow T^{-2}$

Moment of inertia $I = m\ r^2 \rightarrow M\ L^2$

Angular momentum $L = r \times p \rightarrow M\ L^2\ T^{-1}$

Torque $= r \times F$ has the same dimension as energy.

Gradients and frequency

Gradient means change per unit length.

Velocity gradient = v / L → T^{-1} (same as frequency)

Temperature gradient = K / L → K L^{-1}

Frequency = 1/T → T^{-1} Charge = I t → A T

Area = L^2 Volume = L^3 Density = M L^(-3)

Same dimension (T^{-1}) does not mean same quantity.

Every force is still a force

Friction, tension, spring force, weight, buoyancy, electrostatic force,

magnetic force, nuclear force, gravitational force: all $[F] = M\ L\ T^{-2}$.

EMF is not a force. Volt = J/C = $M\ L^2\ T^{(-3)}\ A^{-1}$

Every energy (KE, PE, heat, work function) has $[E] = M\ L^2\ T^{-2}$.

Name can trick you. Dimension follows the definition.

What a unit is

Unit = a reference value. Measurement = n u

Example: 5 m means 5 × the reference 1 m. Also 500 cm.

n u = constant, so n1 u1 = n2 u2

If the unit becomes larger, the number becomes smaller.

How to convert units (write these steps)

In a numerical, write these four lines. Examiner ticks each line.

1. Write the given value with its unit.

2. Replace each unit by the new unit (1 m = 100 cm, 1 kg = 1000 g).

3. Keep the power on the unit: m^2 becomes $(100\ cm)^2$.

4. Multiply and write the final number with the new unit.

n1 u1 = n2 u2 (the actual amount does not change)

Bigger unit → smaller number. If your number grew with a bigger unit, check again.

Rough work / extra work

SI vs CGS you will convert

Quantity	SI	CGS
Mass	kg	g
Length	m	cm
Time	s	s
Force	N = kg m/s²	dyne = g cm/s²
Energy	J = kg m²/s²	erg = g cm²/s²
Power	W = J/s	erg/s
Momentum	kg m/s	g cm/s

1 N = 10^5 dyne. 1 J = 10^7 erg.

Conversion method

Write the quantity with its SI unit, then replace each unit.

1 m = 100 cm, 1 kg = 1000 g, time unit is the same second.

Area 5 m^2 = 5 $(100\ cm)^2$ = 5 x 10^4 cm^2

Powers travel with the unit: m^2 becomes $(100\ cm)^2$, not 100 cm^2.

Do not convert 5 and then forget the power on the unit.

Area 5 m^2 in CGS

If area of an object is 5 m^2, find its value in CGS unit.

Solution: Area 5 m^2 in CGS

Step 1: $5\ m^2 = 5\ (1\ m)^2 = 5\ (100\ cm)^2$

Step 2: $= 5 \times 10^4\ cm^2$

$5 \times 10^4\ cm^2$.

25 m/s in CGS

Convert 25 m/s into CGS unit.

Solution: 25 m/s in CGS

Step 1: 25 m/s = 25 (100 cm / s) = 2500 cm/s

2500 cm/s.

Rough work / extra work

36 km/h into m/s

Convert 36 km/h into m/s.

Solution: 36 km/h into m/s

Step 1: 36 km/h = 36 x (1000 m) / (3600 s) = 10 m/s

Step 2: Shortcut: x km/h = x x (5/18) m/s

10 m/s.

1 newton into dyne

Convert 1 N into dyne.

Solution: 1 newton into dyne

Step 1: $1\ N = 1\ kg\ m / s^2 = (10^3\ g)(10^2\ cm) / s^2$

Step 2: $= 10^5\ g\ cm / s^2 = 10^5$ dyne

$1\ N = 10^5$ dyne.

Rough work / extra work

10 J in CGS

Convert 10 J into CGS unit.

Solution: 10 J in CGS

Step 1: $10\ J = 10\ kg\ m^2 / s^2 = 10\ (10^3\ g)(10^2\ cm)^2 / s^2$

Step 2: = 10 x 10^7 erg = 10^8 erg

10^8 erg. (1 J = 10^7 erg)

KE in SI and CGS

Mass m = 10 kg, velocity 4 m/s. Find KE in SI and in CGS.

Solution: KE in SI and CGS

Step 1: KE = (1/2) m v^2 = (1/2)(10)(16) = 80 J

Step 2: 80 J = 80 x 10^7 erg = 8 x 10^8 erg

80 J or 8 x 10^8 erg.

If the length unit is doubled

If the unit of length becomes double the SI metre, the area 5 m^2 in the new unit is

Solution: If the length unit is doubled

Step 1: n u = constant. New length unit L' = 2 m, so 1 m = (1/2) L'

Step 2: 5 m^2 = 5 $(1/2\ L')^2$ = 5/4 = 1.25 new area units

1.25 (because area scales as L^2).

Rough work / extra work

New unit of force

New system: 1 unit of length = 2 m, mass = 10 kg, time = 4 s. Find the unit of force.

Solution: New unit of force

Step 1: $[F] = M\,L\,T^{-2}$, so 1 new force unit = 10 kg x 2 m / $(4\ s)^2$

Step 2: = 20 / 16 = 1.25 N

1 new force unit = 1.25 N.

10 J in a new system

New units: length 10 m, mass 5 kg, time 2 s. Value of 10 J in this system?

Solution: 10 J in a new system

Step 1: 1 new energy unit = $M\,L^2\,T^{-2}$ = 5 x $(10)^2$ / $(2)^2$ = 125 J

Step 2: n_new = 10 / 125 = 0.08

0.08 new energy units.

Rough work / extra work

Density 5 kg/m^3 in a new system

New system: 1 length unit = x m, 1 mass unit = y kg. Value of density 5 kg/m^3?

Solution: Density 5 kg/m^3 in a new system

Step 1: 1 m = 1/x of the new length, 1 kg = 1/y of the new mass

Step 2: 5 kg/m^3 = 5 (1/y) / (1/x)^3 = (5 x^3 / y) new density units

(5 x^3 / y) new units.

Match the dimensions

Match: A angular momentum, B torque, C stress, D pressure gradient.

1. A-I, B-IV, C-III, D-II
2. A-III, B-I, C-IV, D-II
3. A-II, B-III, C-IV, D-I
4. A-IV, B-II, C-I, D-III

I $[M L^{2} T^{-2}]$ II $[M L^{-2} T^{-2}]$ III $[M L^{2} T^{-1}]$ IV $[M L^{-1} T^{-2}]$

Solution: Match the dimensions

Step 1: $L = r\,p \rightarrow M\,L^{2}\,T^{-1}$ = III. Torque = $r\,F \rightarrow M\,L^{2}\,T^{-2}$ = I.

Step 2: Stress = $F/A \rightarrow M\,L^{-1}\,T^{-2}$ = IV. Pressure gradient = P/L → II.

Answer: A-III, B-I, C-IV, D-II.

Density in a new CGS-like system

Density is 4 g/cm^3. New units: length 10 cm, mass 100 g. New density value is

1. 0.04
2. 0.4
3. 40
4. 400

Solution: Density in a new CGS-like system

Step 1: $4\ g/cm^3 = 4\ (100\ g / 100) / (10\ cm / 10)^3$ wait: 1 new mass = 100 g, 1 new length = 10 cm

Step 2: $n' = 4 \times (1/100) / (1/10)^3 = 4 \times 0.01 \times 1000 = 40$

Answer: 40.

Rough work / extra work

Units 01 lock lines

Seven base quantities. Derived = built from these.

Dimension = powers of M, L, T, … in the formula.

[F] = M L T^{-2}. [W] = [torque] = M L^2 T^{-2}. [P] = M L^2 T^(-3).

n u = constant. Larger unit → smaller number. Powers travel with units.

1 N = 10^5 dyne. 1 J = 10^7 erg. x km/h = x x (5/18) m/s.

Theory lines to copy in the paper

1. Physical quantity = numerical value x unit.
2. A unit must be accepted, unchanging, and of suitable size.
3. SI has seven base units. All other units are derived.
4. Dimension does not change with the system of units. Unit does.
5. Angle is dimensionless but has a unit (radian).
6. Work and torque have the same dimension, not the same meaning.

For 3 marks: definition + formula/example + one extra line.

Practice

1. Write dimensions of velocity, acceleration, force, energy, power, pressure.

2. Convert 2 N into dyne and 5 J into erg.

3. Convert 72 km/h into m/s.

4. New units: L = 2 m, M = 2 kg, T = 2 s. Find 1 new unit of energy in joule.

5. Why can work and torque have the same dimension but remain different?

6. Density 8 g/cm^3. New mass unit 200 g, new length unit 5 cm. Find new density.

Next class: uses of dimensional analysis (checking, converting, guessing formulas).